

3D Shape Formulas for Olympiads

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1 Introduction

This document contains a collection of formulas for the volume and surface area for common three-dimensional geometric solids. It is intended as a quick reference for contest geometry. Throughout this text, let V denote volume and S denote surface area. All measurements are assumed to be in consistent units unless otherwise specified.

1.1 Cube

For a cube of side length a :

$$V = a^3 \tag{1}$$

$$S = 6a^2 \tag{2}$$

1.2 Rectangular Prism

For a rectangular prism of side lengths a , b , and c :

$$V = abc \tag{3}$$

$$S = 2(ab + bc + ca) \tag{4}$$

1.3 General Prism

For a prism of base area B , base perimeter P , and height h :

$$V = Bh \tag{5}$$

$$S = 2B + Ph \tag{6}$$

1.4 Pyramid

For a pyramid of base area B , base perimeter P , height h , and slant height l :

$$V = \frac{1}{3}Bh \tag{7}$$

Surface area of regular pyramids(base is regular, apex is directly above center of base):

$$S = B + \frac{1}{2}Pl \quad (8)$$

Surface area of a general pyramid, where a_i denotes the area of the i th triangular lateral face:

$$S = B + \sum_{i=1}^n a_i \quad (9)$$

1.4.1 Regular Tetrahedron

For a regular tetrahedron of side length a :

$$V = \frac{a^3}{6\sqrt{2}} \quad (10)$$

$$A = \sqrt{3}a^2 \quad (11)$$

1.5 Cylinder

For a cylinder of radius r and height h :

$$V = \pi r^2 h \quad (12)$$

$$S = 2\pi r^2 + 2\pi r h \quad (13)$$

1.6 Cone

For a cone of radius r , height h , and slant height l :

$$V = \frac{1}{3}\pi r^2 h \quad (14)$$

$$S = \pi r^2 + \pi r l \quad (15)$$

1.7 Sphere

For a sphere of radius r :

$$V = \frac{4}{3}\pi r^3 \quad (16)$$

$$S = 4\pi r^2 \quad (17)$$

1.8 Frustum of a Cone

For a frustum of a cone of top radius r_1 , bottom radius r_2 , height h , and slant height l :

$$V = \frac{1}{3}\pi h(r_1^2 + r_2^2 + r_1 r_2) \quad (18)$$

$$S = \pi(r_1^2 + r_2^2) + \pi(r_1 + r_2)l \quad (19)$$

Note that l can be calculated with $\sqrt{h^2 + (r_2 - r_1)^2}$

1.9 Frustum of a Pyramid

For a frustum of a pyramid of bottom base area B_1 , top base area B_2 , and height h :

$$V = \frac{h}{3}(B_1 + B_2 + \sqrt{B_1 B_2}) \quad (20)$$

Let a_i be the bottom base edge length, b_i be the top base edge length, and l_i be the slant height of the i th trapezoidal face. The surface area is then:

$$S = B_1 + B_2 + \frac{1}{2} \sum_{i=1}^n (a_i + b_i) l_i \quad (21)$$

For a regular pyramid of bottom base perimeter P_1 , top base perimeter P_2 , and slant height l :

$$S = B_1 + B_2 + \frac{1}{2}(P_1 + P_2)l \quad (22)$$

1.10 Hemisphere

For a hemisphere of radius r :

$$V = \frac{2}{3}\pi r^3 \quad (23)$$

$$S_{\text{curved}} = 2\pi r^2 \quad (24)$$

$$S_{\text{total}} = 3\pi r^2 \quad (25)$$

1.11 Spherical Cap

For a spherical cap of height h and sphere radius r :

$$V = \frac{1}{3}\pi h^2(3r - h) \quad (26)$$

$$S_{\text{curved}} = 2\pi r h \quad (27)$$

For the total surface area, let a denote the radius of the base. Note that $a = \sqrt{2rh - h^2}$. Then:

$$S_{\text{total}} = 2\pi r h + \pi(2rh - h^2) \quad (28)$$

This can also be written as:

$$S_{\text{total}} = 4\pi r h - \pi h^2 \quad (29)$$